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Central Limit Theorem (CLT)

$X_1, \dots, X_n, \dots$: independent, identical

$\sim$ iid X

μ_X, σ_X^2 $\quad M_X(t) = E[e^{tX}]$ (MGF exists)

$Y_n = X_1 + X_2 + \cdots + X_n \sim$ usually diverges....

$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$

$M_{\bar{X}_n}(t) = M_X(t/n)^n = \left(E\left[1 + \frac{tX}{n} + \frac{t^2X^2}{2!n} + \cdots\right]\right)^n$

$\tilde{X}_i = \frac{X_i - \mu_X}{\sigma_X}$ $\quad \tilde{X} = \frac{X - \mu_X}{\sigma_X}$

$\tilde{Y}_n = \tilde{X}_1 + \cdots + \tilde{X}_n$

$\tilde{\bar{X}}_n = \frac{\tilde{X}_1 + \cdots + \tilde{X}_n}{n}$ $\quad M_{\tilde{\bar{X}}_n}(t) = M_{\tilde{X}}(t/n)^n = \left(E\left[1 + \frac{t\tilde{X}}{n} + \frac{t^2\tilde{X}^2}{2n^2} + \cdots\right]\right)^n$

$\tilde{\bar{X}}_n \xrightarrow{d} 0$ $\quad = \left(1 + \frac{t^2}{2n^2} + \cdots\right)^n \to 1$

$\tilde{W}_n = \frac{\tilde{X}_1 + \cdots + \tilde{X}_n}{\sqrt{n}}$ $\quad M_{\tilde{W}_n}(t) = M_{\tilde{X}}(t/\sqrt{n})^n$

$\tilde{W}_n \xrightarrow{d} N(0,1)$ $\quad = \left(1 + \frac{t^2}{2n} + \cdots\right)^n \to e^{t^2/2}$

$X_1, X_2, \dots, X_n, \dots$ indep

μ_X, σ_X^2

$X_n \xrightarrow{p} Y$ $\quad \Pr(|X_n - Y| > \varepsilon) \to 0$ $\quad \varepsilon > 0$

LLN: $\bar{X}_n \xrightarrow{p} \mu_X$

$X_n \xrightarrow{d} Y$ $\quad F_{X_n}(x) \to F_Y(x)$ x: F_Y continuous

$\Updownarrow$

$M_{X_n}(t) \to M_Y(t)$ pointwise

$X_n \xrightarrow{p} Y \Rightarrow X_n \xrightarrow{d} Y$

CLT-based approximation

$X_1, X_2, \dots, X_n \sim$ iid X $\quad \mu_X, \sigma_X^2$

$X_1 + X_2 + \cdots + X_n \overset{\text{approx}}{\sim} N(n\mu_X, n\sigma_X^2)$

Ex: (1) $X_i \sim$ iid Bernoulli $(1/2)$

$S_n = X_1 + \cdots + X_n \sim \text{Binomial}(n, 1/2)$

$$\Pr\left(S_n > \frac{3n}{4}\right) \overset{\text{Markov}}{<} \frac{2}{3}$$

$\frac{3n}{4}$: $n/2 + n/4$

$$\overset{\text{Chebyshev}}{<} \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

$c > 0$

$$\overset{\text{Chernoff}}{<} \inf_{t>0} \left(\frac{1+e^t}{2}\right)^n \cdot e^{-t\frac{3n}{4}} < e^{-cn}$$

$$S_n \overset{\approx}{\sim} N\left(\frac{n}{2}, \frac{n}{4}\right)$$

$$\Pr\left(S_n > \frac{3n}{4}\right) \approx \Pr\left(N(0,1) > \frac{\sqrt{n}}{2}\right) = 1 - \Phi_Z\left(\frac{\sqrt{n}}{2}\right) = \Phi_Z\left(-\frac{\sqrt{n}}{2}\right)$$

$\frac{S_n - n/2}{\sqrt{n}/2} > \frac{\sqrt{n}}{2}$

Lyapunov CLT

$X_1, X_2, \ldots, X_n, \ldots$ indep

need not be identical

X_i : mean μ_i, variance σ_i^2

$S_n = X_1 + \cdots + X_n$

$$s_n^2 = \sum_{i=1}^n \sigma_i^2$$

$$\delta > 0 \quad \left(\lim_{n\to\infty} \frac{1}{s_n^{2+\delta}} \sum_{i=1}^n E\left[|X_i - \mu_i|^{2+\delta}\right] = 0\right)$$

$(\delta = 1)$

$\delta = 1$: growth of 3rd central moment

Then, $$\frac{S_n - \sum_{i=1}^n \mu_i}{s_n} \xrightarrow{d} N(0,1)$$

Bernoulli case

$X_1, X_2, \ldots, X_n \sim$ indep Bernoulli

$X_i \sim \text{Bernoulli}(p_i)$

Lyapunov CLT, $\delta = 1$

$$s_n^2 = \sum_{i=1}^{n} p_i(1-p_i)$$

$$E\left[|X_i - \mu_i|^3\right] = (1-p_i)\,p_i^3 + p_i(1-p_i)^3$$

$\mu_i = p_i$

$$= p_i(1-p_i)\left[p_i^2 + (1-p_i)^2\right]$$

$$\leq p_i(1-p_i)$$

$$\sum_{i=1}^{n} E\left[|X_i - \mu_i|^3\right] \leq \sum_{i=1}^{n} p_i(1-p_i) = s_n^2$$

Lyapunov condition: $? \leq \dfrac{s_n^2}{s_n^3} = \dfrac{1}{s_n} \to 0$ iff $s_n \to \infty$

So, if $\sum_{i=1}^{n} p_i(1-p_i) \to \infty$, then

$$X_1 + \cdots + X_n \approx \text{Normal}$$